LIGHTWEIGHT AUTOMOTIVE MOUNTING BRACKET

CAD Design, Static FEA, Mesh Convergence and Python Analysis | Drawing AMB-001 | October 2026

1. Project overview

I designed and optimized an L-shaped automotive mounting bracket in SolidWorks and used a Python analysis program to compare seven design iterations. The objective was to reduce mass while satisfying defined strength and stiffness criteria for a simplified static loading case.

2. Loading and analysis method

The bracket was modeled using 6061-T6 aluminum with an assumed yield strength of 275 MPa. I applied a 100 N downward load at the upper hole and modeled fixed supports at the two lower mounting holes. A 2.5 kg electrical module under approximately 4.08 g of effective acceleration is an illustrative equivalent-static interpretation of the load, not a vehicle-specific requirement.

I used a minimum yield-based factor of safety of 3 and a maximum resultant displacement of 0.10 mm as project screening criteria. The Python program organized the SolidWorks results and calculated comparisons, safety factors, and mesh-refinement changes.

3. Main results

Metric	Design 1: baseline	Design 7: final
Mass	219.82 g	111.68 g
Maximum von Mises stress	12.75 MPa	32.37 MPa
Maximum resultant displacement	0.02114 mm	0.09300 mm
Yield-based factor of safety	21.57	8.50

Overall mass reduction: 49.19% (108.14 g). The final design satisfied both project screening criteria for the analyzed static load.

4. Design iterations and optimization

The seven recorded versions represent progressive changes to plate thickness, bracket width, base shape, and material-removal cutouts. The table below uses the values saved in the completed Python project.

Design	Mass (g)	Stress (MPa)	Disp. (mm)	FoS
Design 1	219.82	12.75	0.02114	21.57
Design 2	188.26	20.79	0.04660	13.23
Design 3	167.76	32.88	0.09242	8.36
Design 4	150.62	34.27	0.09819	8.02
Design 5	142.10	31.67	0.08747	8.68
Design 6	124.08	31.94	0.09317	8.61
Design 7	111.68	32.37	0.09300	8.50

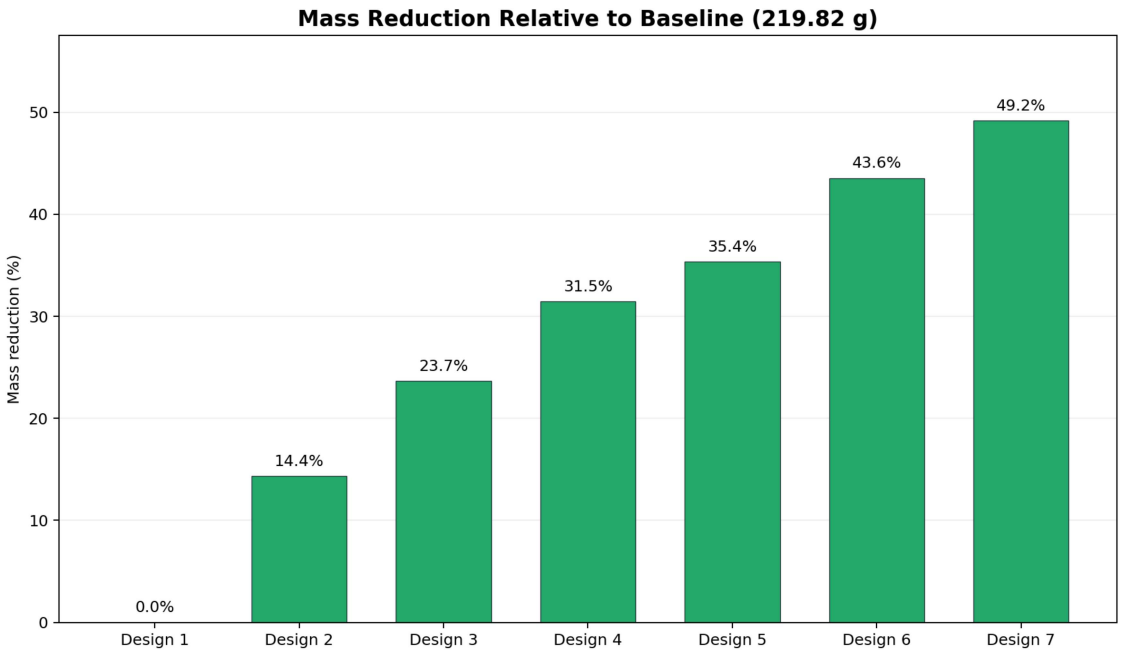


Figure 1. Mass reduction across Designs 1–7, generated from the Python analysis dataset.

Design 7 has the lowest recorded mass. The result reflects a trade-off: reducing material increases stress and displacement relative to the baseline, so the structural limits were checked at every design stage.

5. Structural performance comparisons

The comparison graphs summarize how the recorded maximum stress and displacement vary with bracket mass. These are post-processing plots of SolidWorks simulation outputs, not stresses computed independently by Python.

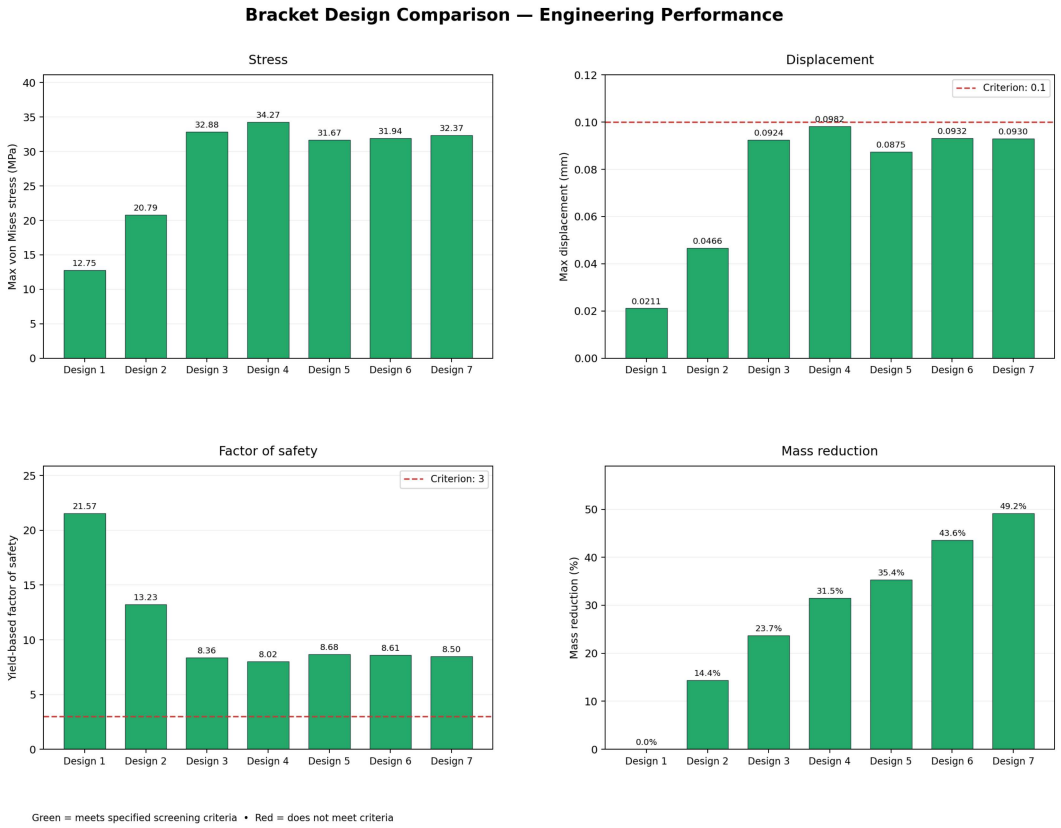


Figure 2. Design comparison of mass, stress, displacement, and safety factor.

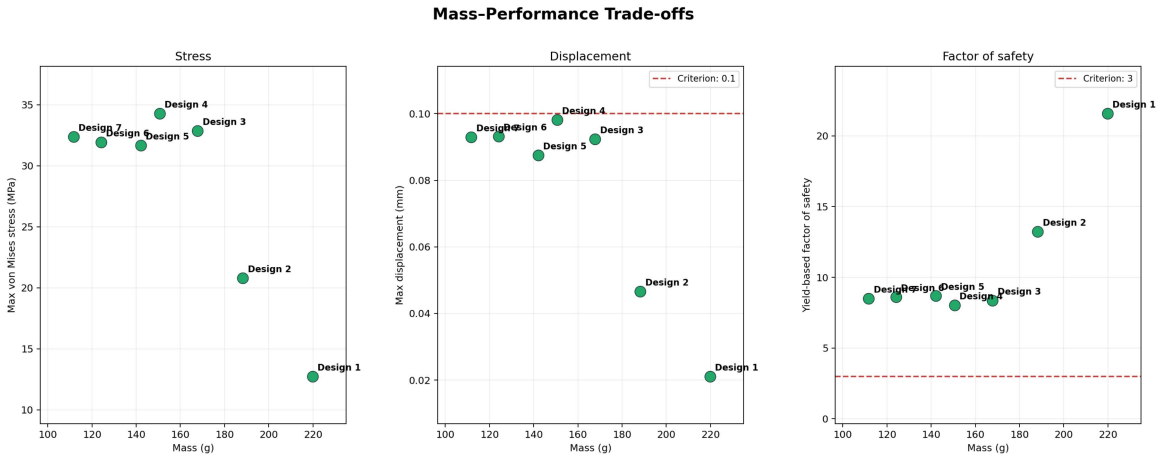


Figure 3. Mass–performance trade-offs for the seven designs.

6. Mesh-refinement analysis

I refined the mesh of Design 7 and compared the predicted maximum von Mises stress and resultant displacement. The finer-mesh results are summarized below.

Mesh	Elements	Nodes	Stress (MPa)	Disp. (mm)
mesh 1	8,704	15,384	25.56	0.09249
Mesh 2	10,925	18,875	25.80	0.09246
Mesh 3	16,747	28,149	26.94	0.09253
Mesh 4	25,304	41,113	28.48	0.09274
Mesh 5	45,878	72,434	31.06	0.09294
Mesh 6	55,843	87,592	31.76	0.09298
final_mesh	64,851	100,892	32.37	0.09300

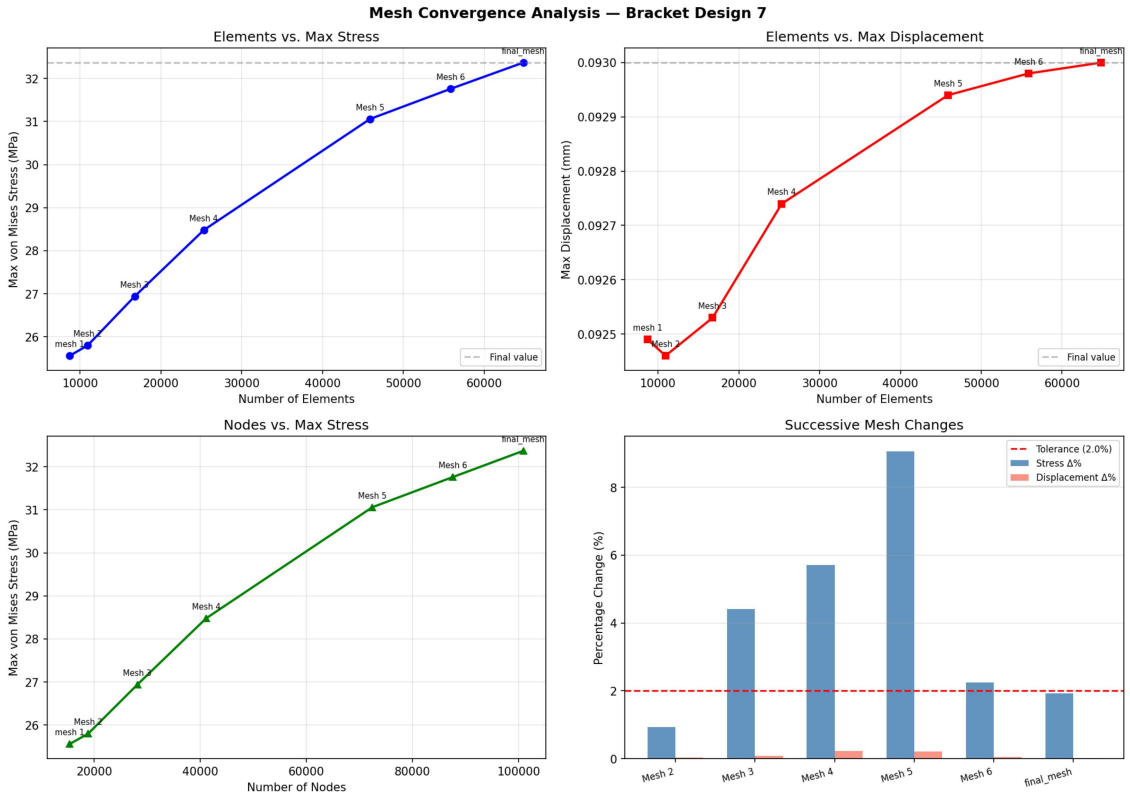


Figure 4. Mesh-refinement trends for the final bracket.

Between the final two mesh levels (55,843 and 64,851 elements), maximum stress changed by **1.92%** and displacement by **0.0215%**. Both changes were below the selected 2% comparison tolerance. The finest mesh result was used for the final design summary.

7. Material comparison

The original SolidWorks study also included a material comparison for the final bracket geometry. These values are retained from the original engineering report as a separate study; they are not part of the seven-design Python dataset.

Material	Mass (g)	Stress (MPa)	Yield (MPa)	FoS
6061-T6	111.68	32.42	275	8.48
7075-T6	116.23	32.36	505	15.61
6063-T6	111.68	32.36	215	6.64
Carbon steel	322.63	33.91	220.6	6.51
Ti-6Al-4V	183.28	32.98	827.4	25.09

I selected 6061-T6 for the main optimization study because its modeled strength met the selected limits while maintaining a low bracket mass. A complete material selection would additionally evaluate fatigue, corrosion, cost, and manufacturing requirements.

8. Final SolidWorks results

The final Design 7 SolidWorks static study produced a maximum von Mises stress of 32.37 MPa and a maximum resultant displacement of 0.09300 mm.

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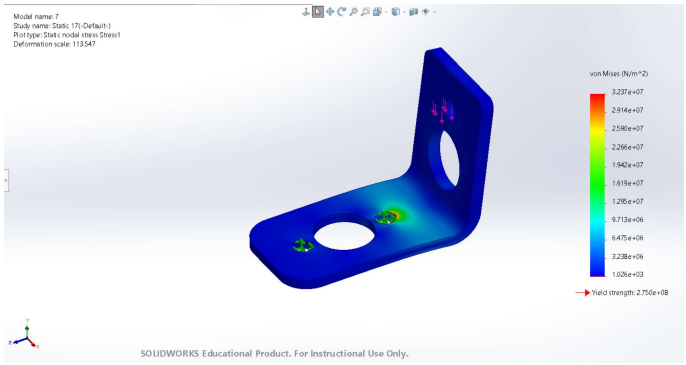


Figure 5. Final Design 7 von Mises stress plot (32.37 MPa).

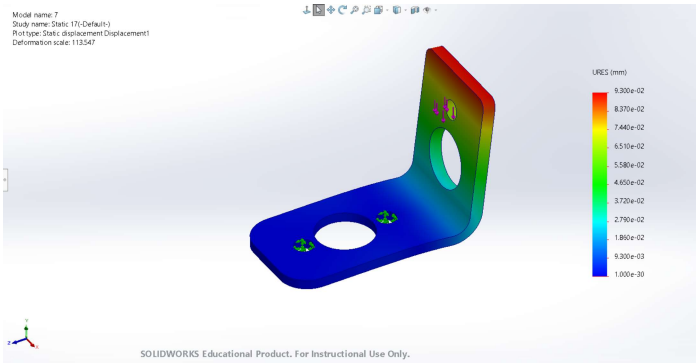


Figure 6. Final Design 7 resultant displacement plot (0.09300 mm).

9. Final bracket engineering drawing

The final drawing documents the bracket's overall geometry, mounting-hole layout, cutouts, and principal dimensions. It is a design-study drawing and does not specify all manufacturing tolerances.

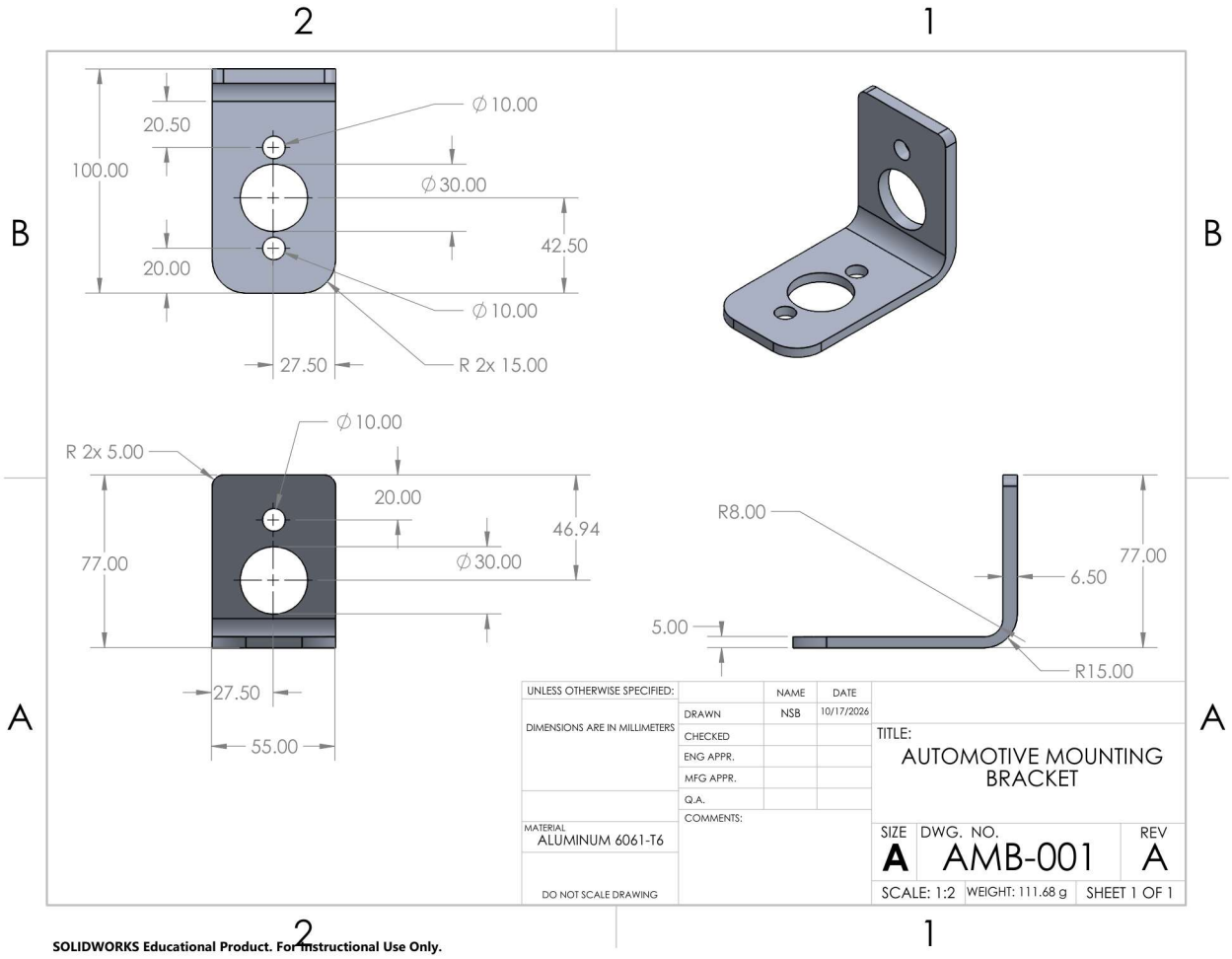


Figure 7. Final SolidWorks engineering drawing, AMB-001 (Revision A).

Final recorded design dimensions include a 100 mm overall length, 55 mm width, 77 mm height, 5 mm horizontal plate thickness, and 6.5 mm upright plate thickness.

10. Python analysis and calculations

I developed a Python tool to store bracket designs and simulation results, calculate yield-based factors of safety, compare design performance, generate engineering graphs, and summarize mesh refinement. The simulation stress and displacement values originated from SolidWorks.

Sample calculations

Mass reduction: (219.82 - 111.68) / 219.82 × 100 = **49.19%**.

Final yield-based factor of safety: 275 / 32.37 = **8.50**.

Final mesh stress change: (32.37 - 31.76) / 31.76 × 100 = **1.92%**.

Final mesh displacement change: (0.09300 - 0.09298) / 0.09298 × 100 = **0.0215%**.

11. Study scope and limitations

This was a linear-static design comparison under one assumed 100 N load case and idealized fixed supports. The selected displacement and factor-of-safety limits were project criteria, not automotive certification standards. The study did not model bolt preload, contact, vibration, fatigue, impact, or physical prototype testing. These effects would be evaluated before any real vehicle application.

12. Conclusion

I reduced the bracket mass from 219.82 g to 111.68 g, a 49.19% reduction. In the final recorded Python dataset, Design 7 had a maximum von Mises stress of 32.37 MPa, maximum displacement of 0.09300 mm, and yield-based factor of safety of 8.50 under the assumed 100 N static load. The project combined iterative SolidWorks CAD design, FEA, mesh refinement, and Python-based engineering analysis.

Data and figure sources

SolidWorks CAD, simulation outputs, and drawing AMB-001 from the original project; Python design and simulation JSON records and generated graphs from the latest project archive. The separate material comparison is reproduced from the earlier engineering report.